

RRC-Raptor-V4H

- I2S

V0.0.1

Contents

i.	VERSION LIST	5
ii.	HASHTAG	6
1.	Overview.....	7
2.	Sound Card Setup	8
3.	Module Setup	9
4.	ALSA (Advanced Linux Sound Architecture)	12
5.	APPENDIX	13

LIST OF FIGURES

FIGURE 1. THE POSITION OF I2C CONNECTOR AND SSI CONNECTOR.....	9
FIGURE 2. PINOUT OF I2C CONNECTOR.....	9
FIGURE 3. PINOUT OF SSI CONNECTOR	9
FIGURE 4. PIN CONNECTIONS BETWEEN V4H-RAPTOR AND DA7212.....	11

LIST OF TABLES

TABLE 1. PIN FUNCTIONS OF I2C CONNECTOR.....	10
TABLE 2. PIN FUNCTIONS OF SSI CONNECTOR	10

i. **VERSION LIST**

Description	Version	Date	Author
I2S	V0.0.1	2024/07/01	HowardChao

ii. HASHTAG

#V4H

#kernel

#BSP

#ubuntu

#I2S

1. Overview

Target : Introduce how to test I2S function on V4H-Raptor.

2. Sound Card Setup

Check the device tree “r8a779g0-raptor.dts” if SUPPORT_DA7212 is defined, and select the frequency correctly to play audio files in different sample rate (44.1 kHz/ 48 kHz).

```
&audio_clklin {
    clock-frequency = <22579200>; /* 44.1kHz groups */
//    clock-frequency = <24576000>; /* 48    kHz groups */
};

&rcar_sound {
    ...
    /* audio_clkout0 */
    #clock-cells = <0>;
    clock-frequency = <11289600>; /* 44.1kHz groups */
//    clock-frequency = <12288000>; /* 48    kHz groups */

    ports {
        rsnd_port0: port@0 {
            rsnd_endpoint: endpoint {
#ifdef SUPPORT_DA7212
                remote-endpoint = <&da7212_endpoint>;
#else
                remote-endpoint = <&lt9611_audio>;
#endif
            }
        }
    }
};
```

Then check the sound card device can be found and listed under “/sys/class/sound/” as shown below.

```
$ ls /sys/class/sound/
card0  controlC0  pcmC0D0c  pcmC0D0p  timer
```


3. Module Setup

The positions of the I2C connector (CN25) and SSI connector (CN28), with their pinouts, are shown in Figures 1, 2, and 3. The pin functions are listed below in Table 1 and Table 2.

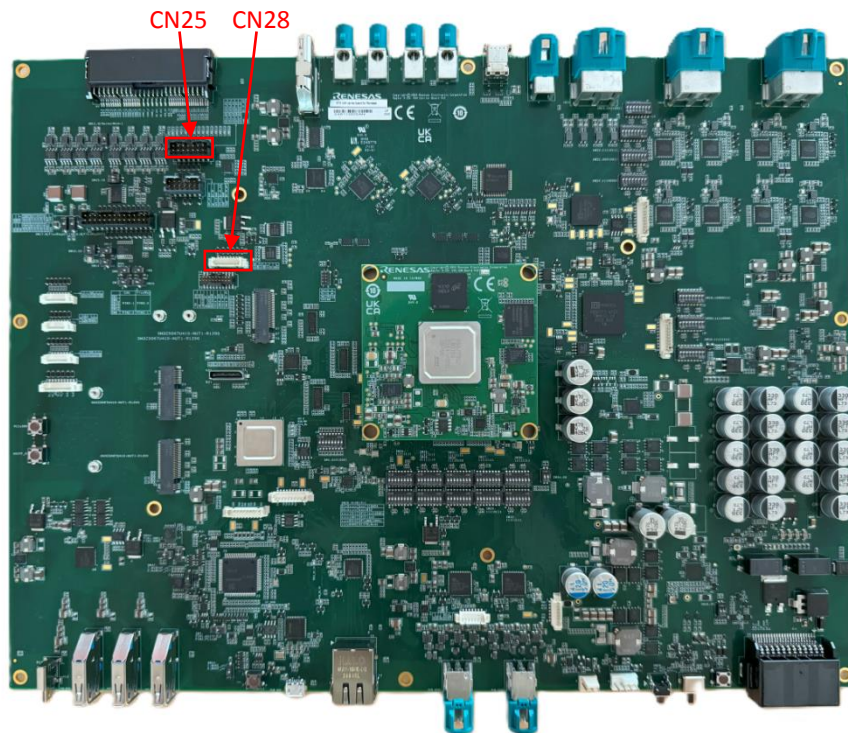


Figure 1. The position of I2C connector and SSI connector



Figure 2. Pinout of I2C connector

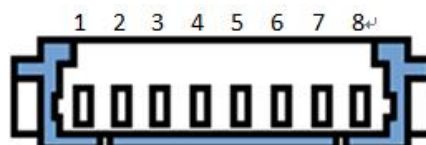


Figure 3. Pinout of SSI connector

Table 1. Pin Functions of I2C connector

Function	PIN	PIN	Function
D3.3V_I2C	1	2	GND
D3.3V_I2C	3	4	GND
I2C0_SCL	5	6	I2C0_SDA
I2C1_SCL	7	8	I2C1_SDA
I2C2_SCL	9	10	I2C2_SDA
I2C3_SCL	11	12	I2C3_SDA
I2C4_SCL	13	14	I2C4_SDA

Table 2. Pin Functions of SSI connector

PIN	Function
1	AUDIO_CLKOUT_V
2	GND
3	AUDIO_CLKIN_V
4	GND
5	SSI_SCK_V
6	SSI_WS_V
7	SSI_SD_V
8	D3.3V_SPI

Confirm the pin connections between V4H-Raptor and the audio module are correct, as shown in Figure 2 below. Pin5 and pin10 of the audio module are used for playback and speaker respectively.

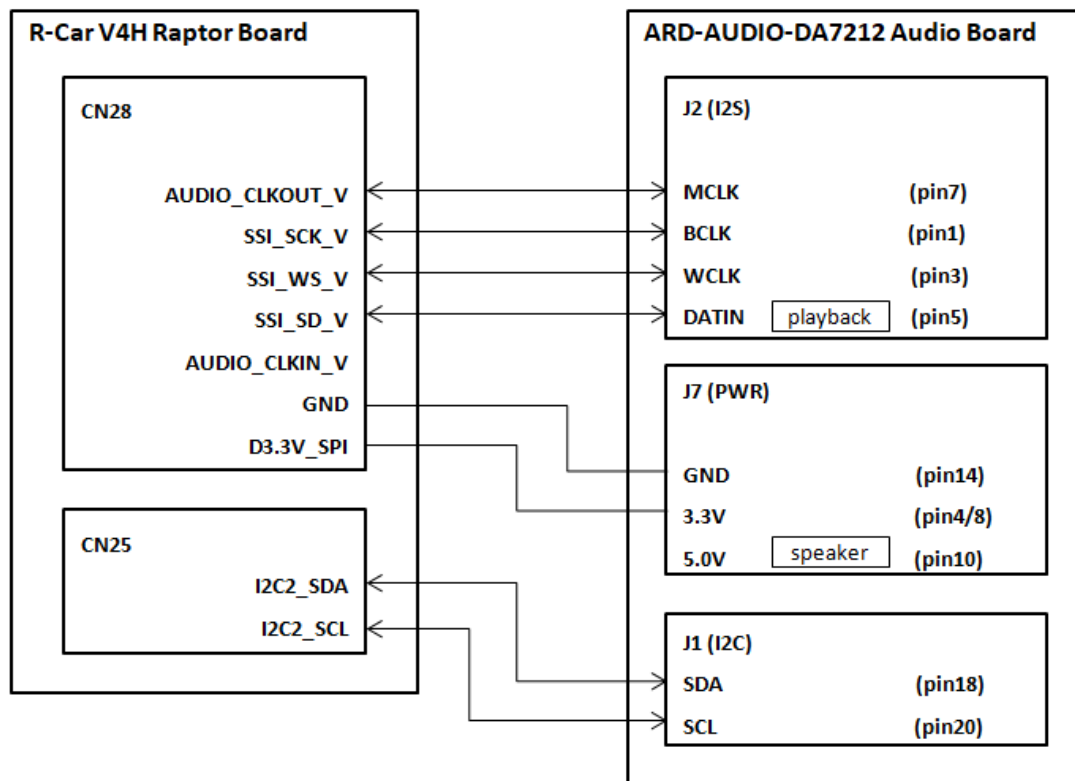


Figure 4. Pin connections between V4H-Raptor and DA7212

4. ALSA (Advanced Linux Sound Architecture)

ALSA is an audio framework and driver system in Linux. It provides a driver interface for audio devices, allowing applications to communicate with and control various sound cards and audio equipment.

Here are some basic commands to use below, or operate “alsamixer” GUI to adjust these parameters.

- Turn on the headphone

```
$ amixer -q cset name='Headphone Volume' 40%  
$ amixer -q cset name='Headphone Switch' on  
$ amixer -q cset name='Mixout Left DAC Left Switch' on  
$ amixer -q cset name='Mixout Right DAC Right Switch' on
```

- Turn off the headphone

```
$ amixer -q cset name='Headphone Switch' off
```

- Turn on/off the speaker

```
$ amixer -q cset name='Lineout Switch' on/off
```

- Lists all the available audio devices detected by ALSA

```
$ aplay -l  
**** List of PLAYBACK Hardware Devices ****  
card 0: I82801AAICH [Intel 82801AA-ICH], device 0: Intel ICH [Intel 82801AA-ICH]  
  Subdevices: 1/1  
  Subdevice #0: subdevice #0
```

- Play a WAV file named filename.wav for 10 seconds with stereo by a specific audio device

```
$ aplay -d 10 -c 2 -D hw:0,0 filename.wav
```

5. APPENDIX